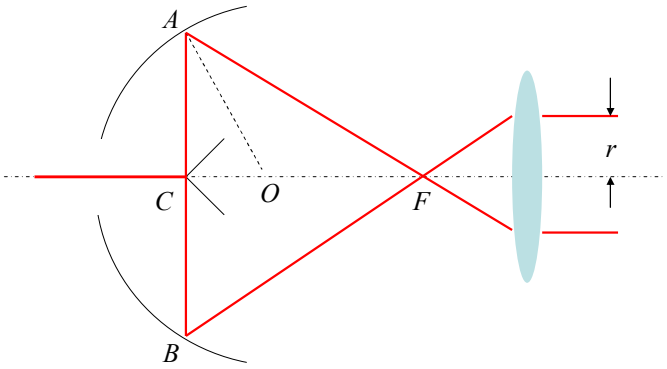
1、



2、 $CO = x$ $\angle CAO = \alpha$

$$\sin\alpha = \frac{x}{R}, \tag{1}$$

$$r = f\cot(2\alpha) \quad . \tag{2}$$

$$x = R\sin(\frac{1}{2}\arctan\frac{f}{r}). \tag{3}$$

5 （3） 5